Set 21: Energy Consumption Analysis

AIM

To analyze energy consumption using R programming by creating a histogram, density plot, scatter plot, and bar chart to understand consumption patterns, temperature relationships, peak hours, and renewable energy usage.

ALGORITHM

1. Create a dataset containing Sector, Region, Month, Temperature, Units Consumed, Cost, Renewable Usage, and Peak Hours.
2. Load the ggplot2 and dplyr libraries.
3. Import or create the energy consumption dataset.
4. Plot a histogram and density curve for Units Consumed.
5. Create a scatter plot between Temperature and Units Consumed, using bubble size for Peak Hours and transparency for clarity.
6. Calculate average Renewable Usage for each Sector.
7. Create a bar chart to compare average renewable energy usage across sectors.
8. Display the plots and interpret the energy consumption patterns.

PROGRAM

```
library(ggplot2)
library(dplyr)


energy <- data.frame(
  Sector = c("Residential", "Commercial", "Industrial",
             "Residential", "Commercial", "Industrial"),
  Region = c("North", "South", "West", "East", "North", "South"),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
)
```

```
ggplot(energy, aes(x = Units_Consumed)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 5, fill = "skyblue", color = "black") +
  geom_density(color = "red", linewidth = 1) +
  labs(title = "Energy Consumption Distribution",
       x = "Units Consumed (kWh)", y = "Density")

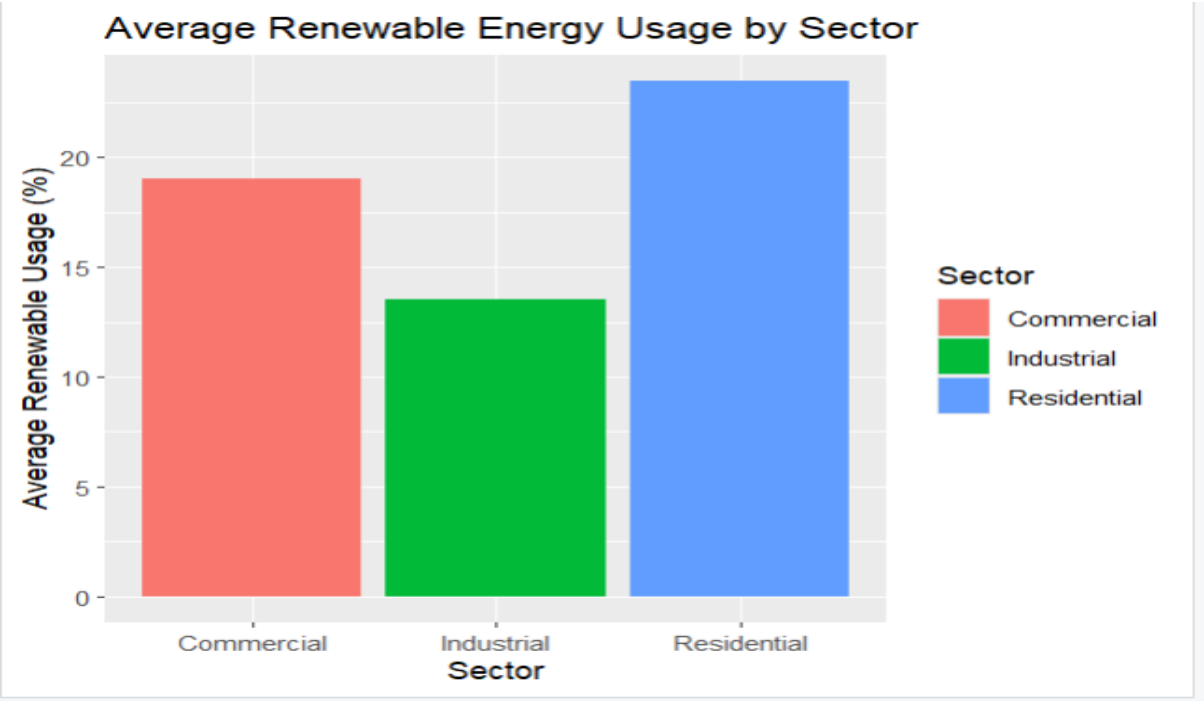
ggplot(energy, aes(x = Temperature, y = Units_Consumed,
                   size = Peak_Hours, color = Sector)) +
  geom_point(alpha = 0.5) +
  labs(title = "Temperature vs Energy Consumption",
       x = "Temperature (°C)", y = "Units Consumed (kWh)",
       size = "Peak Hours")

renewable <- energy %>%
  group_by(Sector) %>%
  summarise(Average_Renewable = mean(Renewable_Usage))

ggplot(renewable, aes(x = Sector, y = Average_Renewable,
                      fill = Sector)) +
  geom_col() +
  labs(title = "Average Renewable Energy Usage by Sector",
       x = "Sector", y = "Average Renewable Usage (%)")

print(energy)
print(renewable)
```

OUTPUT:



RESULT

The histogram and density plot show the distribution of energy consumption. The scatter plot illustrates the relationship between temperature and energy consumption, with bubble sizes representing peak hours. The bar chart compares average renewable energy usage across sectors.

The average renewable usage is:

Sector	**Average renewable usage**
Residential	23.5%
Commercial	19%
Industrial	13.5%

Residential has the highest average renewable usage, while Industrial has the lowest.